

CENTRAL BOARD OF SECONDARY EDUCATION — PRACTICE EXAMINATION
CLASS IX • SCIENCE (SUBJECT CODE: 086) • ACADEMIC SESSION 2026-27

SET F — RAPID FIRE SPEED TEST (30 MINS)

CHAPTER 01: EXPLORATION: ENTERING THE WORLD OF SECONDARY SCIENCE
TIME ALLOWED: 30 MINUTES MAXIMUM MARKS: 25 MARKS SERIES: CBS-IX-SCI-SET-F-01

Student's Name: _____

Roll No.: _____

Class & Section: Class IX - _____

Date of Exam: _____

Invigilator's Sign: _____

GENERAL INSTRUCTIONS (SET F - RAPID FIRE SPEED TEST):

1. The question paper comprises **25 Rapid Fire Multiple Choice Questions** of **1 mark each**.
2. Designed for **Rapid Recall & High-Speed Accuracy**: Recommended target time is **30 Minutes**.
3. All questions are compulsory. There is no negative marking.
4. Darken the corresponding bubble on the **OMR Sheet** completely using a blue/black ballpoint pen.

 **SPEED CHALLENGE INSTRUCTIONS:** Answer promptly without looking at notes. Test your instant recall of formulas, scientific definitions, and core NCERT concepts!

Q.1 What is the SI base unit of thermodynamic temperature?

- (A) Degree Celsius ($^{\circ}\text{C}$) (B) Kelvin (K)
(C) Fahrenheit ($^{\circ}\text{F}$) (D) Joule (J)

Q.2 How many millimeters (mm) are there in 1 kilometer (km)?

- (A) 10^3 mm (B) 10^5 mm
(C) 10^6 mm (D) 10^9 mm

Q.3 Which of the following physical quantities is a scalar quantity?

- (A) Velocity (B) Displacement
(C) Mass (D) Acceleration

Q.4 What is the least count of an ordinary school laboratory meter scale?

- (A) 1 cm (B) 1 mm (0.1 cm)
(C) 0.1 mm (D) 0.01 cm

Q.5 If the effective length of a simple pendulum is made 4 times longer, its time period becomes:

- (A) Half (B) 2 times
(C) 4 times (D) 16 times

Q.6 What is the density of pure liquid water at 4°C in SI units?

- (A) 1 kg/m^3 (B) 100 kg/m^3
(C) 1000 kg/m^3 (D) 10000 kg/m^3

Q.7 Which physical quantity has the SI unit $\text{kg} \cdot \text{m}^{-3}$?

- (A) Pressure (B) Density
(C) Force (D) Momentum

Q.8 What is the time period of a seconds pendulum?

- | | |
|----------------|-----------------|
| (A) 1.0 second | (B) 2.0 seconds |
| (C) 0.5 second | (D) 4.0 seconds |

Q.9 Which laboratory apparatus is used to accurately measure a fixed volume of 25.0 mL of liquid?

- | | |
|-------------------|------------------------|
| (A) Beaker | (B) Volumetric Pipette |
| (C) Conical flask | (D) Test tube |

Q.10 How many significant figures are in the measurement 0.05020 m?

- | | |
|-------|-------|
| (A) 2 | (B) 3 |
| (C) 4 | (D) 5 |

Q.11 If an object sinks in water, its relative density is:

- | | |
|--------------------|-----------------|
| (A) Equal to 1 | (B) Less than 1 |
| (C) Greater than 1 | (D) Zero |

Q.12 The pitch of a micrometer screw gauge is 1 mm and it has 100 circular divisions. Its least count is:

- | | |
|--------------|-------------|
| (A) 0.1 mm | (B) 0.01 mm |
| (C) 0.001 mm | (D) 0.05 mm |

Q.13 Which flame color indicates complete combustion in a properly aerated Bunsen burner?

- | | |
|---------------------------|-----------------------------|
| (A) Yellow luminous flame | (B) Non-luminous blue flame |
| (C) Green flame | (D) Smoky orange flame |

Q.14 What is the value of standard atmospheric pressure at sea level in Pascals (Pa)?

- | | |
|------------------------------------|------------------------------------|
| (A) $1.013 \times 10^5 \text{ Pa}$ | (B) $1.013 \times 10^3 \text{ Pa}$ |
| (C) 100 Pa | (D) 9.8 Pa |

Q.15 Which quantity remains constant during a fair controlled scientific experiment?

- | | |
|--------------------------|------------------------|
| (A) Independent variable | (B) Dependent variable |
| (C) Controlled variables | (D) Random variable |

Q.16 Convert 25°C to the Kelvin scale:

- | | |
|-----------|-----------|
| (A) 248 K | (B) 298 K |
| (C) 300 K | (D) 273 K |

Q.17 What is the volume of a cube having side length 2 cm?

- | | |
|----------------------|-----------------------|
| (A) 4 cm^3 | (B) 6 cm^3 |
| (C) 8 cm^3 | (D) 16 cm^3 |

Q.18 What does a beam balance compare?

- | | |
|-------------|--------------------------|
| (A) Weights | (B) Gravitational masses |
| (C) Volumes | (D) Densities |

Q.19 Parallax error is classified under which category of experimental errors?

- | | |
|-----------------------------|----------------------------------|
| (A) Instrumental zero error | (B) Personal observational error |
| (C) Environmental error | (D) Theoretical error |

Q.20 What is the symbol for the SI prefix 'micro'?

- (A) \$m\$ (B) \$M\$
(C) \$\mu\$ (D) \$n\$

Q.21 A piece of wood of volume 50 cm^3 and density 0.8 g/cm^3 has mass:

- (A) 40 g (B) 62.5 g
(C) 50 g (D) 0.016 g

Q.22 Which unit is a derived SI unit?

- (A) Meter (B) Second
(C) Newton (D) Kilogram

Q.23 What is the acceleration due to gravity on the Moon compared to Earth?

- (A) Identical (B) Approx. $\frac{1}{6}$ of Earth
(C) Approx. 6 times Earth (D) Zero

Q.24 Which instrument is used to determine the specific gravity (relative density) of liquids directly?

- (A) Barometer (B) Hydrometer
(C) Manometer (D) Anemometer

Q.25 If 10 vernier divisions equal 9 main scale divisions (1 MSD = 1 mm), what is the vernier constant?

- (A) 0.1 mm (B) 0.01 mm
(C) 1.0 mm (D) 0.9 mm

CBSE CANDIDATE OMR ANSWER RESPONSE GRID (SET F)

01 (A) (B) (C) (D)	02 (A) (B) (C) (D)	03 (A) (B) (C) (D)	04 (A) (B) (C) (D)	05 (A) (B) (C) (D)
06 (A) (B) (C) (D)	07 (A) (B) (C) (D)	08 (A) (B) (C) (D)	09 (A) (B) (C) (D)	10 (A) (B) (C) (D)
11 (A) (B) (C) (D)	12 (A) (B) (C) (D)	13 (A) (B) (C) (D)	14 (A) (B) (C) (D)	15 (A) (B) (C) (D)
16 (A) (B) (C) (D)	17 (A) (B) (C) (D)	18 (A) (B) (C) (D)	19 (A) (B) (C) (D)	20 (A) (B) (C) (D)
21 (A) (B) (C) (D)	22 (A) (B) (C) (D)	23 (A) (B) (C) (D)	24 (A) (B) (C) (D)	25 (A) (B) (C) (D)

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & RAPID EXPLANATIONS (SET F)

Q. No.	Correct	Scientific Rationale & Formula Summary
Q.1	(B)	Kelvin (K) is the SI base unit of temperature. 0°C corresponds to 273.15 K.
Q.2	(C)	$1\text{ km} = 1000\text{ m} = 1000 \times 1000\text{ mm} = 10^6\text{ mm}$.
Q.3	(C)	Mass has only magnitude with no directional dependence, making it a scalar quantity.
Q.4	(B)	A standard school meter rule is graduated in millimeters; its smallest division is 1 mm (0.1 cm).
Q.5	(B)	$T = 2\pi\sqrt{L/g} \propto \sqrt{L}$. If length increases by 4 times, T increases by $\sqrt{4} = 2$ times.
Q.6	(C)	Water has maximum density at 4°C: $1.00\text{ g/cm}^3 = 1000\text{ kg/m}^3$.
Q.7	(B)	Density is mass per unit volume ($\text{kg/m}^3 = \text{kg} \cdot \text{m}^{-3}$).
Q.8	(B)	A seconds pendulum takes 1 second to swing from one extreme to the other; its full oscillation time period is exactly 2.0 seconds.
Q.9	(B)	A volumetric pipette has a calibration mark delivering an exact volume (25.0 mL) with high precision.
Q.10	(C)	Leading zeros (0.0) are placeholders. '5', '0', '2', and the trailing '0' are significant (4 significant figures).
Q.11	(C)	An object sinks when its density exceeds the density of water (Relative density > 1).
Q.12	(B)	$\text{Least count} = \frac{1\text{ mm}}{100} = 0.01\text{ mm}$.
Q.13	(B)	When the air collar is open, ample oxygen produces a roaring, non-luminous blue flame of complete combustion.
Q.14	(A)	$1\text{ atm} = 760\text{ mm of Hg} = 1.013 \times 10^5\text{ Pa}$ (approx. 101.3 kPa).
Q.15	(C)	Controlled variables are kept strictly constant so that only the independent variable affects the dependent variable.
Q.16	(B)	$T(K) = 25 + 273 = 298\text{ K}$.
Q.17	(C)	$\text{Volume} = s^3 = (2\text{ cm})^3 = 8\text{ cm}^3$.
Q.18	(B)	A beam balance compares an unknown gravitational mass against standard reference masses ($m_1\text{ g} = m_2\text{ g} \Rightarrow m_1 = m_2$).
Q.19	(B)	Parallax is a personal observational error caused by the observer's line of sight not being perpendicular to the scale.
Q.20	(C)	Micro is denoted by the Greek letter μ (mu) and represents 10^{-6} .
Q.21	(A)	$\text{Mass} = \text{Volume} \times \text{Density} = 50\text{ cm}^3 \times 0.8\text{ g/cm}^3 = 40\text{ g}$.
Q.22	(C)	Newton ($\text{kg} \cdot \text{m/s}^2$) is a derived unit of force. Meter, second, and kilogram are base units.
Q.23	(B)	Surface gravity on the Moon is about 1.63 m/s^2 , which is approximately $1/6$ of Earth's 9.8 m/s^2 .
Q.24	(B)	A hydrometer floats in liquids to directly measure relative density based on its depth of immersion.
Q.25	(A)	$\text{VC} = 1\text{ MSD} - 1\text{ VSD} = 1\text{ mm} - 0.9\text{ mm} = 0.1\text{ mm}$.